Policy 5 on validation seeds disjoint from its training

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Exploratory, outside every family. Question, seeds and reading rule declared in `DECISIONS.md` at commit **5a15a35**, before any episode (Filipe, 2026-11-08: "Package B").

Summary

Question	Answer
Does any choice for training seed 5 use 95001 to 95003?	No. Xu, Li and the DQN keep their final state, Auad's theta_best is chosen on gate seed 97005, and the shared Auad configuration was chosen with policy 1
Reading rule	Policy 5 reads as **unaffected** by the overlap: Xu et al. W1 gap -0.067 (policies 1 to 4: -0.222 to -0.096), p90 gap -1.153 (-0.864 to -0.711); Li et al. W1 gap 0.006 (policies 1 to 4: -0.037 to 0.010), p90 gap -0.258 (-0.242 to -0.191); Deep Q-network W1 gap 0.053 (policies 1 to 4: -0.017 to 0.016), p90 gap -0.359 (-0.392 to -0.308); Auad et al. W1 gap -0.024 (policies 1 to 4: -0.012 to 0.039), p90 gap -0.549 (-0.530 to -0.327)
Policy 5's curve on 95501 to 95503	Largest move of W1 from its peak-25 value: Xu et al. 0.120 min (peak 50); Li et al. 0.024 min (peak 100); Deep Q-network 0.084 min (peak 175); Auad et al. 0.000 min (peak 50)
Where 35.9 min comes from	The calibration run of 2026-11-02 (normal supply, seeds 201 to 210): mean of per-episode p90 35.896, p90 of pooled orders 35.896; the 35.8 of Results is the mean of per-episode p90 on seeds 1001 to 1150 (35.828)

Policy 5 reads as **unaffected** by the overlap under the declared rule: for every model, its gap lies within the range of policies 1 to 4, widened by 0.10 min for W1 and 0.5 min for p90.

Before the widening, policy 5's gap lies outside the range of policies 1 to 4 in 6 of 8 checks; a fifth value drawn like the other four falls outside their range with probability 2/5, about 3.2 of 8. Of these, 4 lie below the range (policy 5 relatively lower on 95001 to 95003, its training seeds) and 2 above it; the distances are Xu et al. W1 0.030 (above), Xu et al. p90 0.289 (below), Li et al. p90 0.016 (below), Deep Q-network W1 0.037 (above), Auad et al. W1 0.012 (below), Auad et al. p90 0.019 (below) min.

1. What the code keeps, and which seeds its choices used

Model	Kept state and seeds
Xu et al.	Keeps the final value table, fitted once on the nearest-courier transitions of all 200 peaks; training seed 5's peaks ran on episode seeds 95000 to 95199, and no choice reads any validation seed (the curve is recorded, never used).
Li et al.	Keeps the actor after peak 200 (final); training seed 5 trains on 95000 to 95199 with a temperature schedule fixed by the peak count, and no choice reads any validation seed.
Deep Q-network	Keeps the network after peak 200 (final); training seed 5 trains on 95000 to 95199 with an epsilon schedule fixed by the peak count, and no choice reads any validation seed.
Auad et al.	Keeps theta_best, the theta with the lowest cost on the gate peak, which for training seed 5 is seed 97005 (big-cluster start, normal supply); training seed 5 trains on 95000 to 95199; the configuration (alpha 1e-7, m 3,000, n 1,000) was chosen once from training seed 1 (seeds 91000 to 91199) by its cost on 95001 to 95003 and holds for all five policies, so no policy-5 state ran in any choice on those seeds.
Jusup et al.	Keeps the epoch with the best validation objective on 95001 to 95003 (mean-field rollouts from those seeds' start positions); training seed 5 trains on seeds 103000 to 103726 (98000 + 1000 k + epoch, 727 epochs), disjoint from the validation seeds, so it is not affected and does not run here.

No choice for training seed 5 used 95001 to 95003, so the check ran.

2. What ran

- 288 episodes on FE-SIM-1 (`sim/policy5_validation.py`, 20 workers, one BLAS thread per process): the curve code's validation cases (big-cluster and scattered starts, normal supply, fitted behavior) on seeds 95501 to 95503, which no training, gate or choice of these four models ever ran.
- Policy 5 at every checkpoint (25 to 200 peaks): the recorded checkpoint files of Li, the DQN and Auad; Xu's tables refit from the recorded transitions (the refit at 200 peaks equals the kept table bit for bit). Policies 1 to 4 at the kept state. Each episode is the call the training code's own evaluation makes (a test checks it).
- The rows for 95001 to 95003 are the existing curve rows. At one BLAS thread the current code reproduces them: the budget check's peak-200 rows of policy 1 equal them bit for bit.

3.1 Kept state, mean over the six cases (minutes; Auad's cost in its own units)

Model	Policy	W1, 95001-95003	W1, 95501-95503	p90, 95001-95003	p90, 95501-95503	Auad cost, 95001-95003	Auad cost, 95501-95503
Xu et al.	1	8.604	8.701	47.00	47.86		
Xu et al.	2	8.665	8.769	47.89	48.64		
Xu et al.	3	8.507	8.729	47.54	48.25		
Xu et al.	4	8.635	8.757	47.57	48.36		
Xu et al.	5	8.669	8.736	46.39	47.54		
Li et al.	1	7.784	7.790	39.73	39.97		
Li et al.	2	7.810	7.846	39.68	39.87		
Li et al.	3	7.813	7.803	39.71	39.92		
Li et al.	4	7.828	7.829	39.69	39.91		
Li et al.	5	7.809	7.803	39.66	39.92		
Deep Q-network	1	8.384	8.400	39.59	39.96		
Deep Q-network	2	8.403	8.387	39.63	39.94		
Deep Q-network	3	8.388	8.388	39.59	39.98		
Deep Q-network	4	8.430	8.432	39.59	39.95		
Deep Q-network	5	8.406	8.353	39.62	39.98		
Auad et al.	1	8.455	8.415	40.14	40.66	88,916.6	96,413.0
Auad et al.	2	8.544	8.509	40.46	40.79	122,906.2	129,858.7
Auad et al.	3	8.468	8.479	40.18	40.71	89,185.5	97,014.2
Auad et al.	4	8.467	8.442	40.27	40.67	90,072.3	97,358.1
Auad et al.	5	8.470	8.494	40.14	40.69	88,691.4	95,908.3

3.2 Gap, 95001-95003 minus 95501-95503, at the kept state

Model	Metric	Policy 5	Policies 1 to 4	Beyond that range	Widened range	Within
Xu et al.	W1	-0.067	-0.222 to -0.096	0.030 (above)	-0.322 to 0.004	yes
Xu et al.	p90	-1.153	-0.864 to -0.711	0.289 (below)	-1.364 to -0.211	yes
Li et al.	W1	0.006	-0.037 to 0.010	0	-0.137 to 0.110	yes
Li et al.	p90	-0.258	-0.242 to -0.191	0.016 (below)	-0.742 to 0.309	yes
Deep Q-network	W1	0.053	-0.017 to 0.016	0.037 (above)	-0.117 to 0.116	yes
Deep Q-network	p90	-0.359	-0.392 to -0.308	0	-0.892 to 0.192	yes
Auad et al.	W1	-0.024	-0.012 to 0.039	0.012 (below)	-0.112 to 0.139	yes
Auad et al.	p90	-0.549	-0.530 to -0.327	0.019 (below)	-1.030 to 0.173	yes
Auad et al.	Auad cost	-7,216.9	-7,828.7 to -6,952.5	0		no rule (descriptive)

3.3 Policy 5's curve (mean over the six cases)

Model	Peak	W1, 95501-95503	W1, 95001-95003	Gap	p90, 95501-95503	p90, 95001-95003	Auad cost, 95501-95503
Xu et al.	25	8.786	8.752	-0.034	46.65	46.21	
Xu et al.	50	8.666	8.649	-0.016	46.83	46.10	
Xu et al.	75	8.747	8.613	-0.134	47.13	46.24	
Xu et al.	100	8.806	8.671	-0.135	47.25	46.32	
Xu et al.	125	8.698	8.654	-0.044	47.81	46.69	
Xu et al.	150	8.706	8.667	-0.038	47.31	46.39	
Xu et al.	175	8.734	8.708	-0.026	47.67	46.51	
Xu et al.	200	8.736	8.669	-0.067	47.54	46.39	
Li et al.	25	7.805	7.822	0.017	40.01	39.90	
Li et al.	50	7.822	7.848	0.025	39.98	39.77	
Li et al.	75	7.802	7.777	-0.025	39.90	39.70	
Li et al.	100	7.829	7.835	0.006	39.91	39.74	
Li et al.	125	7.811	7.781	-0.030	39.93	39.73	
Li et al.	150	7.810	7.844	0.034	39.90	39.68	
Li et al.	175	7.814	7.778	-0.036	39.92	39.75	
Li et al.	200	7.803	7.809	0.006	39.92	39.66	
Deep Q-network	25	8.314	8.293	-0.020	39.96	39.70	
Deep Q-network	50	8.330	8.353	0.023	39.95	39.61	
Deep Q-network	75	8.377	8.322	-0.054	40.04	39.67	
Deep Q-network	100	8.355	8.359	0.004	39.96	39.65	
Deep Q-network	125	8.347	8.337	-0.010	40.01	39.63	
Deep Q-network	150	8.388	8.357	-0.031	39.98	39.63	
Deep Q-network	175	8.398	8.383	-0.014	39.97	39.61	

Model	Peak	W1, 95501-95503	W1, 95001-95003	Gap	p90, 95501-95503	p90, 95001-95003	Auad cost, 95501-95503
Deep Q-network	200	8.353	8.406	0.053	39.98	39.62	
Auad et al.	25	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	50	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	75	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	100	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	125	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	150	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	175	8.494	8.470	-0.024	40.69	40.14	95,908.3
Auad et al.	200	8.494	8.470	-0.024	40.69	40.14	95,908.3

Checkpoints 25 to 200 give identical rows for Auad et al.: they hold the same state on both seed sets.

Largest move of W1 from the peak-25 value, peaks 50 to 200:

Model	95501-95503	At peak	95001-95003	At peak
Xu et al.	0.120	50	0.139	75
Li et al.	0.024	100	0.045	75
Deep Q-network	0.084	175	0.112	200
Auad et al.	0.000	50	0.000	50

4. The 35.9 min of the threats paragraph

Source	p90 order to door (min)	Runs	Seeds
calibration, mean of per-episode p90	35.90	calibration `avaliacao/20261102-1026-calibration` (code 572610b): normal supply, 10 episodes, nearest idle courier, calibration's scattered start, every courier full-time and staying where it is while idle	201 to 210
calibration, p90 of pooled orders	35.90	calibration `avaliacao/20261102-1026-calibration` (code 572610b): normal supply, 10 episodes, nearest idle courier, calibration's scattered start, every courier full-time and staying where it is while idle	201 to 210
study runs, mean of per-episode p90	35.83	frozen code, nearest courier, scattered start, normal supply, fitted behavior, 150 episodes	1001 to 1150
Meituan orders delivered alone, p90 of pooled orders	41.08	Meituan single-order waves, 17:00 to 20:59	

The 35.9 min is the calibration run's p90 at normal supply, the number the calibration report prints beside Meituan's 41.1. Both statistics give it there: the mean of the ten per-episode p90 values and the p90 of the pooled orders round to 35.9. Meituan's 41.1 is the p90 of pooled orders, so the like-for-like figure is the pooled one. The 35.8 of Results is another sample: the study's frozen code (fitted idle behavior, part-time couriers, the scenario start), seeds 1001 to 1150, mean of per-episode p90.

Files

`p5_kept.csv`, `p5_gap.csv`, `p5_curve.csv`, `p5_curve_move.csv`, `p5_p90_sources.csv`; run rows in `rows.jsonl` beside `run.json`.

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